Reg.No : ……

Name : ……

MAHATHMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

THIRD SEMESTER

DISCIPLINE SPECIFIC CORE COURSE (DSC A) - MG3DSCBTG201- CELL BIOLOGY AND GENETICS

(2024 ADMISSION ONWARDS)

Duration: 1½ Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) andAppreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

One word Answer

Answer any **10** questions

(Each question carries **2** marks)

1. Name two major proteins found in ECM and state their function. **[K]/ [CO1]**
2. Mention the types of crossing over? **[U]/ [CO3]**
3. Write any two functions of nucleus? **[U]/ [CO2]**
4. What are laws of genetics? **[K]/ [CO3]**
5. List any two factors that can alter gene frequency in a population. **[K]/ [CO3]**
6. Which are the protein involved in cell cycle regulation? **[U]/ [CO2]**
7. Name the check points involved in regulation? **[K]/ [CO2]**
8. Mention two roles of the extracellular matrix in animal cells. **[K]/ [CO1]**
9. Name two transport mechanisms in plasma membrane. **[U]/ [CO1]**
10. Name any two cell junctions. **[K]/ [CO2]**
11. Which are the organelles involved in protein synthesis. **[K]/ [CO2]**
12. What is called the observable characteristics of an individual organism? With example. **[U]/ [CO3]**

(10×2= 20)

Part B

Short Essay

Answer any **4** questions

(Each question carries **5** marks)

13. Describe the structure and components of the extracellular matrix. **[K]/ [CO1]**
14. Explain the models of the plasma membrane. **[K]/ [CO1]**
15. Explain nucleus and its functions. **[U]/ [CO2]**
16. Describe the phases of the cell cycle with a diagram. **[K]/ [CO2]**
17. Explain Mendel's laws of inheritance with suitable examples. **[U]/ [CO3]**
18. Define mutation and explain its types. **[K]/ [CO3]**
19. Explain genetic equilibrium with Hardy-Weinberg principle. **[U]/ [CO3]**

(5×4**= 20)**

Part C

Essay

Answer any **1** question

(Each question carries **10** marks)

20. Explain the molecular regulation of the cycle and the importance of checkpoints. **[K]/ [CO1]**

21. Explain linkage and crossing over. **[U]/ [CO3]**

(1×10= 10)